

NATURAL LANGUAGE PROCESSING

(Computer Science & Engineering)

Time: 3 hours

Max Marks: 60

All questions carry equal marks

Answer the following:**05 x 12 = 60 Marks**

1. a) Design a simple architecture for a voice assistant. Explain how each part help in understanding user input. CO1 L4 6 M
b) Explain sentence structure, phrase structure rules, and the role of syntax in natural language understanding. CO1 L2 6 M
OR
2. a) Explain different levels of Language Analysis. CO1 L2 6 M
b) Explain the major applications of NLP. CO1 L2 6 M
3. a) What is Transition Network Grammar (TNG)? Explain their structure and role in syntactic parsing. CO2 L2 6 M
b) Define grammar based parser for chatbot that uses morphology features, and entropy scoring. Explain each part. CO2 L2 6 M
OR
4. a) Define Entropy and Cross Entropy. Explain their significance in evaluating language models. CO2 L2 6 M
b) Explain the process of top-down parsing vs bottom-up parsing by analyzing how each parser would handle the same sentence. CO2 L2 6 M
5. a) How are questions handled in Context Free Grammars (CFGs)? Explain the techniques used to represent question formation in CFGs. CO3 L2 6 M
b) Discuss about human preferences in parsing. CO3 L4 6 M
OR
6. a) Compare shift-reduce parsers and deterministic parsers. CO3 L4 6 M
b) Explain the importance of robust syntactic parsers in language-learning applications. CO3 L2 6 M
7. a) Discuss about the types of language models. CO4 L4 6 M
b) Design a basic auto complete system that uses semantic interpretation and n-gram modeling. Explain the work flow. CO4 L3 6 M
OR
8. a) Explain how language modeling and semantic interpretation are used in text prediction systems. CO4 L2 6 M
b) What is an n-gram model and how does it help in predicting the next word? Explain with an example. CO4 L2 6 M
9. a) Describe the possible approaches to Machine Translation. CO5 L2 6 M
b) Discuss about the structure of the Anusaraka system. CO5 L2 6 M
OR
10. a) What is meant by giving up agreement in Anusaraka output? Explain why this decision is taken. CO5 L1 6 M
b) Explain how cross-Lingual IR works, when the work is in one language and the documents are in another. CO5 L2 6 M
